

Implementing US Instant Payment Returns with the Form3 API

Audience: Software engineers integrating Form3 US Instant into banking platforms.

Outcome: By the end of this tutorial, you will have successfully submitted a return for a previously received US Instant payment using the Form3 API.

1. Requirements

You should have:

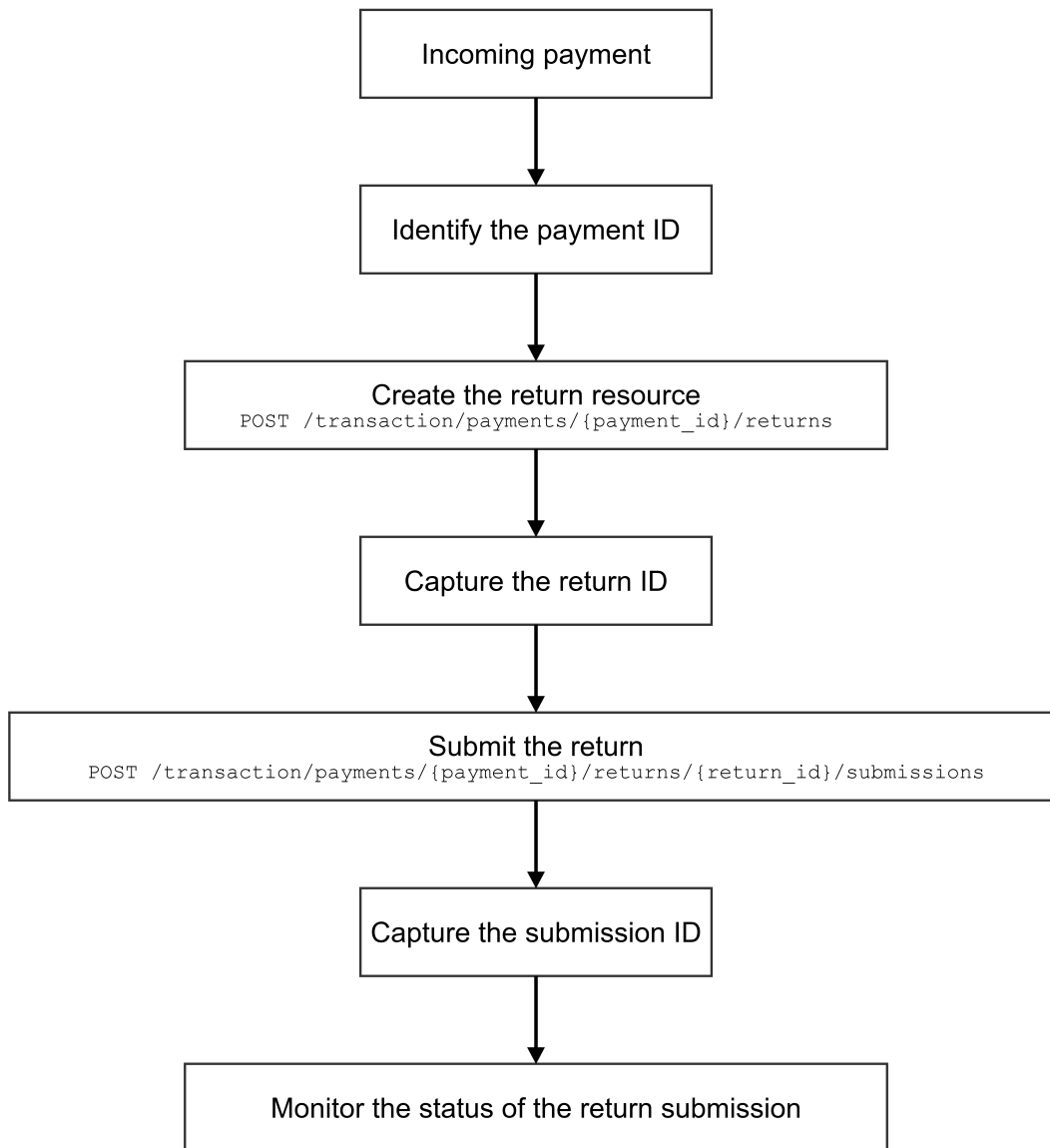
- Access to the Form3 API environment.
- Your organisation ID, a private key available to your application, and its corresponding public key registered with Form3.
- A payment that your organization has **already received**.
- The payment ID of the received payment.

Note on Payment Returns

A payment return is created **against an existing payment**. You cannot submit a return unless the original payment already exists and has been received by your organization.

2. How Payment Returns Work

A payment return instructs Form3 to return funds from a previously received payment back to the sending institution.



Unlike sending a new payment, you do not create an entirely new payment resource. Instead, you create a **return resource** that is associated with the original payment, and then submit a **return submission** to initiate the return process.

Note on Payment Schemes

Form3's US Instant product provides a unified interface for both the RTP and FedNow schemes. This tutorial applies to both, as they share the same API endpoints and message structures.

3. Creating a Return Resource

To create a return resource, you will need to locate the original payment and submit a `POST` request to the returns endpoint for that payment.

3.1 Create the Return Resource Request

Submit a `POST` request to the payment returns endpoint for the payment you want to return by calling the `/payments/{id}/returns (US Instant)` endpoint and providing the payment ID in the request path.

```
POST /transaction/payments/{payment_id}/returns
```

```
Accept: application/json
```

```
Content-Type: application/json
```

```
Date: <request_date>
```

```
Digest: <request_body_digest>
```

```
Authorization: <generated_signature_header>
```

If Form3 accepts the request and creates the resource, the API returns `201 Created`. The response contains the newly created Return resource, including its client-generated ID in `data.id`. Your integration will need this identifier to submit the return.

3.2 Example Return Resource Request

Request completeness

This tutorial supplies every field shown in the Form3 Postman example for this endpoint. The published endpoint reference does not currently identify mandatory and optional request fields individually.

```
{
  "data": {
    "id": "f6d61d86-6bd0-49c0-a278-c4fd0a0b9dc3",
    "type": "returns",
    "organisation_id": "<organization_id>",
    "attributes": {
      "amount": "600.00",
      "currency": "USD",
      "return_code": "AC02"
    }
  }
}
```

3.3 Return Resource Request Fields

The request example includes the fields in the table below. The endpoint reference does not explicitly label individual request fields as required, so confirm the current validation rules against your Form3 API specification before implementing production validation.

Field	Location	Value used in this tutorial	Description
payment_id	Path	Existing payment UUID	Identifies the received payment to return.
data.id	Body	Client-generated UUID	Uniquely identifies the return resource.
data.type	Body	returns	Identifies the resource type.
data.organisation_id	Body	Your organisation UUID	Identifies the Form3 organisation creating the return.
data.attributes.amount	Body	600.00	Specifies the amount to return.
data.attributes.currency	Body	USD	Specifies the return currency.
data.attributes.return_code	Body	AC02	Specifies the reason for returning the payment.

4. Submitting the Return Submission Resource

After creating the return resource, you must submit it to initiate the return process by sending a `POST` request to the return submissions endpoint.

4.1 Create the Return Submission Request

Submit a `POST` request to the return submissions endpoint for the return resource you just created by calling the `/payments/{id}/returns/{return_id}/submissions` endpoint.

POST /transaction/payments/{payment_id}/returns/{return_id}/submissions

Content-Type: application/json

Accept: application/json

Date: <request_date>

Digest: <request_body_digest>

Authorization: <generated_signature_header>

If Form3 accepts the request and creates the resource, the API returns 201 Created . The response contains the newly created return submission resource, including its client-generated ID in `data.id` .

A 201 Created response confirms that Form3 created the Return Submission resource. It does not confirm final scheme delivery. Monitor the submission resource's status to track subsequent processing.

4.2 Example Return Submission Resource Request

```
{
  "data": {
    "id": "8a8faaf1-bff4-4b78-9f12-e366d68654e9",
    "type": "returnsubmissions",
    "organisation_id": "<organization_id>"
  }
}
```

4.3 Return Submission Request Fields

Field	Location	Value used in this tutorial	Description
<code>data.id</code>	Body	Client-generated UUID	Uniquely identifies the return submission resource.
<code>data.type</code>	Body	<code>returnsubmissions</code>	Identifies the resource type.
<code>data.organisation_id</code>	Body	Your organisation UUID	Identifies the Form3 organisation creating the return.

5. Python Example

The following Python example illustrates how to create and submit a payment return using the `requests` library.

This example is simplified for clarity and does not include application-specific exception handling, retries or logging.

The example assumes that your application provides a `create_form3_signed_headers()` helper implemented according to Form3's HTTP Request Signing specification. Form3, Python and the `requests` library do not provide this function.

```

import json
import uuid

import requests

def create_form3_signed_headers(method, url, body, private_key, public_key_id):
    # Implement Form3 HTTP Request Signing here.
    # Return the request headers, including:
    # - Accept
    # - Content-Type
    # - Date
    # - Digest
    # - Authorization
    #
    # The signature calculation must use the exact body supplied
    # through the body argument.
    raise NotImplementedError(
        "Implement Form3 HTTP Request Signing before running this example."
    )

base_url = "https://<form3-api-host>/v1"
payment_id = "your-actual-payment-uuid-here"
private_key = "your_private_key_here"
public_key_id = "your_public_key_id_here"
organization_id = "your_organization_id_here"

# Create the return resource.
return_id = str(uuid.uuid4())

return_payload = {
    "data": {
        "id": return_id,
        "type": "returns",
        "organisation_id": organization_id,
        "attributes": {
            "amount": "600.00",
            "currency": "USD",
            "return_code": "AC02",
        },
    },
}

```

```

return_url = f"{base_url}/transaction/payments/{payment_id}/returns"

return_body = json.dumps(return_payload, separators=(",", ":"))

return_headers = create_form3_signed_headers(
    method="POST",
    url=return_url,
    body=return_body,
    private_key=private_key,
    public_key_id=public_key_id,
)

response = requests.post(
    return_url,
    headers=return_headers,
    data=return_body,
)

response.raise_for_status()

return_id = response.json()["data"]["id"]
print(f"Return created successfully with ID: {return_id}")

# Submit the return.
submission_id = str(uuid.uuid4())

submission_payload = {
    "data": {
        "id": submission_id,
        "type": "returnsubmissions",
        "organisation_id": organization_id,
    }
}

submission_url = (
    f"{base_url}/transaction/payments/{payment_id}/returns/{return_id}/submissions"
)

submission_body = json.dumps(submission_payload, separators=(",", ":"))

submission_headers = create_form3_signed_headers(
    method="POST",
    url=submission_url,

```

```
        body=submission_body,
        private_key=private_key,
        public_key_id=public_key_id,
    )

    response = requests.post(
        submission_url,
        headers=submission_headers,
        data=submission_body,
    )
    response.raise_for_status()

    submission_id = response.json()["data"]["id"]
    print(f"Return submitted successfully with ID: {submission_id}")
```

6. Troubleshooting

The most common issues when creating a return are either a rejected request or a return of the wrong payment.

6.1 API Request Rejection

Check that:

- the request matches the API specification
- the payment exists
- the payment was received through your Form3 organization, and you are using its correct `payment_id`
- the payment has already been received
- the request body contains a valid `organisation_id`
- the currency is `usd`
- the return code is valid for your use case

6.2 Incorrect Payment Returned

Because the return endpoint operates directly on an existing payment resource, always verify the `payment_id` before issuing the request. A common operational safeguard is to retrieve and display payment details before allowing an operator or workflow to submit the return.

7. Integration Testing

Before promoting your integration, verify that you can:

- create a return for a known received payment
- receive `201 Created` when creating the return
- extract and store the returned `data.id`
- create a return submission using the return ID
- receive `201 Created` for the submission
- extract and store the submission's `data.id`
- handle non-successful responses without attempting to parse success data
- verify that the submission's initial creation does not get treated as final scheme confirmation
- monitor or retrieve the Return Submission resource to observe its processing status

8. Security Considerations

When implementing payment returns, ensure that your application:

- verifies that operators are authorized to initiate returns
- protects API credentials used for payment operations
- records an audit trail for every submitted return
- avoids exposing payment identifiers in logs where unnecessary

9. Related Resources

- [POST /payments/{id}/returns \(US Instant\)](#)
- [POST /payments/{id}/returns/{return_id}/submissions](#)
- [Payment Returns](#)
- [Return a Payment with ARUCS](#)
- [Send a Payment](#)
- [Send a Signed Request](#)

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